

Silver Commodity Price Forecasting & Market Intelligence Dashboard

- Subtitle: End-to-End Machine Learning Analytics & Power BI Visualization
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- Target Audience: Evaluation Panel / Zetheta Review

Executive Summary

- Project Purpose:** Designed an institutional-grade analytics command center to forecast silver spot prices and evaluate macro risks.
- Core Stack:** Python (Scikit-Learn, XGBoost) for predictive feature engineering and Power BI for interactive visualization.
- Key Outcome:** Merged multi-frequency data (Daily OHLCV, Monthly Macroeconomics, and Futures Curves) into a unified star schema.

Problem Statement & Objectives

- **Market Volatility:** Precious metals experience high volatility driven by currency shifts, global crises, and investor sentiment.

- **Core Objectives:**

- Build an automated machine learning pipeline to predict directional price movements

- (Ensemble_Pred) .

- Construct a multi-tab business intelligence dashboard for risk monitoring and historical trend analysis.

Data Architecture & Star Schema

- **Dataset Sources:**

- *Daily OHLCV*: Historical spot pricing, daily returns, and trading volume (silver_daily_ohlc_v_2000_2025).

- *Macroeconomic Data*: US Dollar Index (DXY) and Gold spot prices (silver_macro_economic_monthly).

- *Futures Term Structure*: Contract months and basis calculations (silver_futures_contracts).

- **Modeling Design**: Established a clean star schema in Power BI to optimize cross-filtering and measure performance.

Machine Learning Pipeline (Python Backend)

- Feature Engineering:** Extracted technical indicators, moving averages, daily percentage returns, and lag features to optimize signal detection.
- Ensemble Modeling:** Trained robust **Random Forest** and **XGBoost** algorithms to generate predictive pricing structures.
- Integration:** Exported synthesized predictions (Ensemble_Pred) directly into the data model for real-time comparison against actual market behavior.

Power BI Dashboard – Tab 1 (Executive Overview)

- **KPI Command Center:** Real-time tracking of latest spot prices and daily return percentages.
- **Historical Trajectory:** Time-series visualizations capturing structural market movements from 2000 through the 2011 peak and post-2020 market recovery.
- **Volume Dynamics:** Highlighting liquidity surges that precede significant market breakouts.

Power BI Dashboard – Tab 2 (Macro & Futures Intelligence)

- **Gold-Silver Correlation:** Scatter plots mapping silver performance against gold pricing benchmarks.
- **Macroeconomic Overlays:** Tracking the inverse relationship between silver prices and the strength of the US Dollar Index (DXY)
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- **Futures Term Structure:** Ribbon charts analyzing market contango and backwardation phases using the calculated basis.

Power BI Dashboard – Tab 3 (Crisis Backtesting)

- **Custom DAX Dimensions:** Engineered a Crisis_Period calculated column to segment historical data across major economic shocks.
- **Stress-Testing Windows:** Evaluated market behavior and model performance
 - during the **2008 Financial Crisis**, the **2011 Silver Peak**, and the **2020 COVID-19 Crash**.
- **Risk Evaluation:** Benchmarked returns and trading volumes across high-stress market environments.

Key Insights & Findings

- **Macro Drivers:** Confirmed a strong structural inverse correlation between silver returns and DXY Index fluctuations.
- **Predictive Efficacy:** Ensemble models accurately captured directional trends during high-volatility regime shifts.
- **Liquidity Insights:** Volume spikes serve as reliable early indicators for institutional accumulation phases.

Conclusion & Future Enhancements

- **Business Impact:** Delivers a fully scalable, automated risk-monitoring tool for commodity analysts and traders

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- **Future Scope:**

- Integration of real-time API feeds for live data ingestion.

- Deployment of Natural Language Processing (NLP) models for automated news sentiment tracking.